

RECOLLECTIONS ON DG-CATEGORIES

ABSTRACT. Notes on AusCat talk by Ross Street, taken by Steve Lack

Motivations: biographical note for Academy on Max; Ganter-Kapranov paper on 2-dimensional trace.

DG-cats: invented independently by Max (“complex categories”) and Eilenberg-Moore.

For homology, might think you could just take a differential object (A, d) in $\mathbf{Ab}$ (or some abelian category). Now define BA to be the image and ZA the kernel of d ; $HA := ZA/BA$, the homology of A .

Can’t make this monoidal in a nice way (what is $(A, d) \otimes (B, d)$?)

A *graded abelian group* is a family $(A_n)_{n \in \mathbb{Z}}$ of groups indexed by $\mathbb{Z}$. A *differential graded abelian group* is a graded abelian group with homomorphisms $d : A_n \rightarrow A_{n-1}$ with $d^2 = 0$.

This is a (pointed) functor category so it is Grothendieck abelian. Furthermore, you now get a monoidal structure

$$(A \otimes B)_n = \sum_{p+q=n} A_p \otimes B_q$$

with $d(a \otimes b) = da + (-1)^p a \otimes db$ where $a \in A_p, b \in B_q$. This gives a symmetric monoidal closed category $\mathbf{DGAAb}$.

Have an embedding $\mathbf{Ab} \rightarrow \mathbf{DGAAb}$ sending $M \in \mathbf{Ab}$ to the complex concentrated in degree 0. The image of $\mathbb{Z}$ under this embedding is the unit.

There is only one symmetry:

$$c_{A,B} : A \otimes B \cong B \otimes A : a \otimes b \mapsto (-1)^{pq} b \otimes a.$$

In fact $\mathbf{DGAAb}^{\geq 0}$ is equivalent to the category of internal ω -categories in $\mathbf{Ab}$ (here $\mathbf{Ab}$ could be replaced by any Cauchy complete additive category). Closely related to Dold-Kan theorem: $\mathbf{DGAAb}^{\geq 0} \simeq [\Delta^{\text{op}}, \mathbf{Ab}]$.

Joyal: Dold-Kan is a fancy version of Newton’s theorem about recapturing a list of quantities from the iterated first differences.

Internal hom:

$$[A, B]_n = \prod_p [A_p, B_{p+n}]$$

with $d(f) = d \circ f - (-1)^n f \circ d$. An f with $df = 0$ is called a *chain map of degree n*. So the chain maps of degree 0 are the morphisms of $\mathbf{DGAAb}$.

Now a *differential graded category* is a category enriched in $\mathbf{DGAAb}$.

$\mathbf{DGAAb}$ is an abelian category. What are the projective objects? It turns out that $A \in \mathbf{DGAAb}$ is projective iff each A_n is projective and $H_n A = 0$. This means that we can only get projective resolutions of finite length for acyclic chain complexes (zero

homology). Given projective resolution of length 1

$$0 \longrightarrow Q \longrightarrow P \longrightarrow A \longrightarrow 0$$

have $HA = 0$ (by long exact sequence argument).

Eilenberg-Moore introduced “relative homological algebra”, probably to explain the last chapter of Cartan-Eilenberg — possibly this is what led to Kelly work on factorization systems. We would like A to be “projective” when all of A_n , $B_n A$, $Z_n A$, and $H_n A$ are projective. So we have a restricted notion of epimorphism corresponding to these projectives: these are the chain maps inducing epis on B , Z , H .

Very useful short exact sequences

$$0 \longrightarrow Z_n A \longrightarrow A_n \longrightarrow B_{n-1} A \longrightarrow 0$$

$$0 \longrightarrow B_n A \longrightarrow Z_n A \longrightarrow H_n A \longrightarrow 0$$

Lawvere has promoted the picture (“unity and identity of adjoint opposites”)

$$\begin{array}{ccc} & \text{DGA}b & \\ -/B \swarrow & \uparrow & \searrow Z \\ & \text{GA}b & \end{array}$$

with HA as image of the induced $ZA \rightarrow A/BA$.

Cauchy completion What is the Cauchy completion of a DG-category? For any $\mathcal{V}$ and $\mathcal{V}$ -category $\mathcal{A}$, write $Q\mathcal{A} \subseteq [\mathcal{A}^{\text{op}}, \mathcal{V}]$ for the closure of $\mathcal{A}$ under absolute colimits.

Let’s think about a DG-category $\mathcal{A}$. Then $\mathcal{A}(A, B)$ is a chain complex. Call $f \in \mathcal{A}(A, B)_n$ a *protomap* of degree n . Call it a chain map of degree n when $df = 0$. The underlying category $\mathcal{A}_0$ has the same objects as $\mathcal{A}$, and chain maps of degree 0.

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & \xleftarrow{v} & & \xleftarrow{u} & \\ & & & & \end{array}$$

where f, g are chain maps of degree m, n ; and u, v are protomaps, with $vf = 1_A$, $gu = 1_C$, $fv + ug = 1_B$. Then B is not necessarily the direct sum of A and C in $\mathcal{A}_0$. But the whole system is preserved by any DG-functor. In fact $v \circ d(u) \in Z_{-m-n-1}\mathcal{A}(C, A)$, and the homology class $\delta_{f,g} = [v \circ d(u)] \in H\mathcal{A}(C, A)$ is called the *deviation* of f and g and is independent of u and v . Now get

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{\delta} A$$

“ B is like the mapping cone of δ ”.

Some constructions in DGAb**.** The *suspension* Σ :

$$(\Sigma A)_n = A_{n-1}, \quad d^{\Sigma A} = -d^A$$

Given chain map $f : A \rightarrow B$ of degree 0, the *mapping cone* of f is the chain complex $Cn(f)$

$$Cn(f)_n = B_n \oplus A_{n-1}, \quad d = \begin{pmatrix} d & f \\ 0 & -d \end{pmatrix}$$

Then have

$$B \begin{array}{c} \xrightarrow{i} \\ \xleftarrow{\dots\dots\dots} \end{array} Cn(f) \begin{array}{c} \xrightarrow{q} \\ \xleftarrow{\dots\dots\dots} \end{array} \Sigma A$$

Also, suppose we have $u : A \rightarrow A$ a chain map of degree 0 with $u^2 = 0$. Let A^u be A as a graded object with $d^u = d + u$. (Tentatively call this a *deformation*.) This is once again absolute. Does not seem to follow from mapping cones, suspensions, and direct sums.

All these constructions (direct sums, suspensions, mapping cones, deformations) are preserved by enriched representables, so can be defined as weighted limits in any DG-category. They are absolute (co)limits.

We have $Cn(f) = (B \oplus \Sigma A) \begin{pmatrix} 0 & f \\ 0 & 0 \end{pmatrix}$ so mapping cones follow from the others.

$Q\mathcal{A}$ certainly includes direct sums, suspensions, and deformations, and retracts. This *could* be all. But the reason this is not obvious was discussed a little by looking at adjoint modules.

Remark added 25 April 2010: The deformation construction can be defined more generally for any protomap $u : A \rightarrow A$ of degree -1 satisfying $u^2 + d(u) = 0$. I am still not sure whether this extra is all we need for the Cauchy completion.